CHENAB RIVER · MARALA HEADWORKS · PAKISTAN

Our flood forecast models

An easy guide to what we tried, why we tried it, how each model makes forecasts, what the saved tests say, and which data went into training.

10-day forecasts | 15-day forecasts | 30-day experiments

Read this first

These are retrospective tests using observed Marala discharge and, in many experiments, reanalysis weather as a stand-in for future forecasts. The 30-day tests do not prove real-time 30-day forecast skill. “Flood” in the comparisons means flow above a training-data 95th-percentile threshold; it is not an official warning category.

Prepared from the model folders and evaluation reports saved in this project. Scores are transcribed from those reports; no models were retrained for these notes.

How to read the scores

Flow error

RMSE is the typical size of forecast errors, with extra penalty for big misses. Lower is better. It is measured here in cubic metres per second (m³/s).

NSE compares a forecast with a simple average-flow reference. 1 is perfect; 0 is about as good as that reference; below 0 is worse. It is **not a percent accuracy score**.

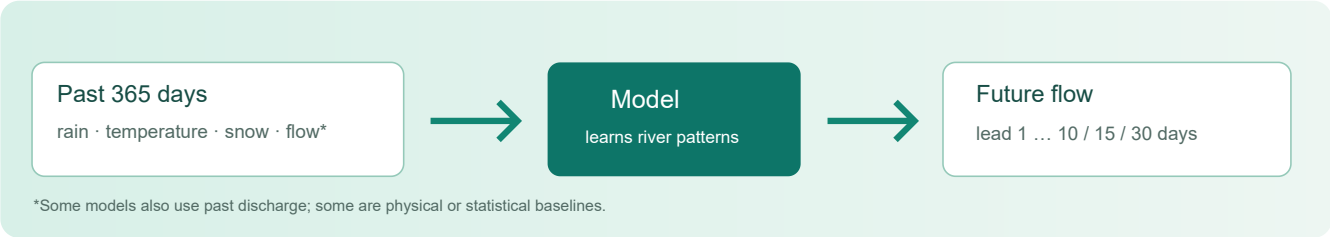
Flood detection

Recall asks: of the observed high-flow days, how many did the model catch? **Precision** asks: of the days it warned, how many were high flow?

There are far fewer high-flow days than ordinary-flow days. So overall accuracy can look high even when many floods are missed. Check flood recall and flood RMSE.

Important limit: most weather-driven tests use realized ERA5-Land weather with added noise, not archived weather forecasts that would have been available on the forecast date. Treat the scores as model comparisons under a weather proxy, not operational warning accuracy. The test years (mainly 2022–2024) have also been viewed in earlier project work.

Forecast picture



“Lead 10” means the flow forecast for ten days after the issue date. Leads are evaluated against observed discharge only; missing targets are excluded.

Section 1 · Model families and why we tried them

We tested simple references first, then machine-learning, transfer-learning, and hydrology-based approaches. The purpose is to see what adds value for Marala, where local observed history is limited and floods are uncommon.

10-day model folder: 19 saved model/run folders

Model	Why it fits / how it predicts	What the saved evidence says
Original hindcast–forecast LSTM	Sequence model for weather-to-flow timing. Reads the previous year of daily inputs, then decodes future weather leads into flows.	Day-10 high-flow RMSE 745.6; recall 0.588 in the common comparison.
Gradient boosting	Strong small-data baseline. Many shallow decision trees combine weather and recent-history summaries.	Day-10 high-flow RMSE 811.5; recall 0.314.
LSTM + gradient-boosting ensemble	Combines a sequence learner with tree predictions to reduce reliance on one model's errors.	Day-10 high-flow RMSE 748.3; recall 0.529.
CAMELS-US 8-basin pretrained LSTM	Starts from patterns learned at eight observed US catchments; tests transfer to a data-limited basin.	Unadapted donor is a transfer diagnostic; not a strong Marala forecast. See fine-tuned result below.
CAMELS-US pretrained, fine-tuned on Marala	Reuses donor weights, then adjusts them using Marala training years.	Day-10 high-flow RMSE 643.0; recall 0.765. Better flood detection in this one test, with many false alarms.
Two-input LSTM trained from scratch	Uses precipitation and temperature only. A control for whether donor pretraining helps.	Day-10 high-flow RMSE 657.6; recall 0.745.
Global Caravan pretrained LSTM, fine-tuned	Transfers a wider multi-basin learned representation; adapted to Marala.	Day-10 high-flow RMSE 706.5; recall 0.647. Pilot fine-tuning gain over matched scratch was small overall and absent for monsoon NSE.
Global LSTM zero-shot / donor-only references	Shows what transferred weights do before local adaptation.	Very poor day-10 flood scores; diagnostic only, not a locally trained candidate.
Snow/water-bucket + LSTM	Simple storage buckets represent snow and catchment water before an LSTM learns remaining patterns.	Day-10 high-flow RMSE 730.2; recall 0.627.
Spatial elevation-band LSTM	Separate elevation bands preserve the fact that high and low parts of the basin receive different rain/snow.	Day-10 high-flow RMSE 896.7; recall 0.196 in the main comparison.
Catchment-mean LSTM control	Same broad weather idea, averaged over the catchment; checks if spatial detail helps.	Day-10 high-flow RMSE 703.6; recall 0.667. Beat the elevation-band version on this split.
15-day encoder–decoder, fine-tuned	Encoder reads history; decoder produces a longer sequence. Fine-tuning reuses donor weights.	Day-10 flood RMSE 649.8; recall 0.804. It also forecasts through day 15.
15-day encoder–decoder, scratch	Same sequence-to-sequence approach, learned only from local training data; transfer control.	Day-10 flood RMSE 673.4; recall 0.804.
Calibrated physical HBV	Water-balance equations turn rain, temperature and evaporation into runoff; parameters are calibrated to Marala.	30-day day-30 flood recall 0.314 in the later experiment; see comparison section.
HBV + LSTM residual	HBV gives a physically based first answer; LSTM learns a correction to its errors.	30-day flood recall 0.745 (best catch rate in that 30-day table), but false alarms and ordinary-flow error matter too.
HBV + LightGBM residual	Same physical starting point; tree boosting corrects the remaining errors.	30-day flood recall 0.529.
Discharge-history-only LSTM	Tests how much yesterday's and recent river flow alone can tell us, without weather.	Older 15-day evaluation; different period/cohort, so not ranked with the main day-10 test.
LSTM + lagged GRRR, gaps retained	Tests a lagged external flow product as an input while leaving target gaps missing.	15-day historic test; not directly comparable with newer 2022–24 scores.
Low-weight GRRR-filled-target LSTM **(rejected)**	Experimental target filling was tested to see if extra pseudo-targets helped learning.	Rejected after evaluation. Do not treat filled values as observed or use this run as a preferred model.

Other saved 10-day entries include persistence and climatology benchmarks, delta-HBV, HBV residual variants, N-HiTS, N-BEATSx, and matched-input controls. Some names above group the same method appearing in multiple horizon folders. See Appendix inventory for exact folder lists.

Section 1 · 30-day model families

These longer-horizon runs were done one at a time. Longer forecasts have less direct day-to-day information, so comparing against simple seasonal flow patterns is especially important.

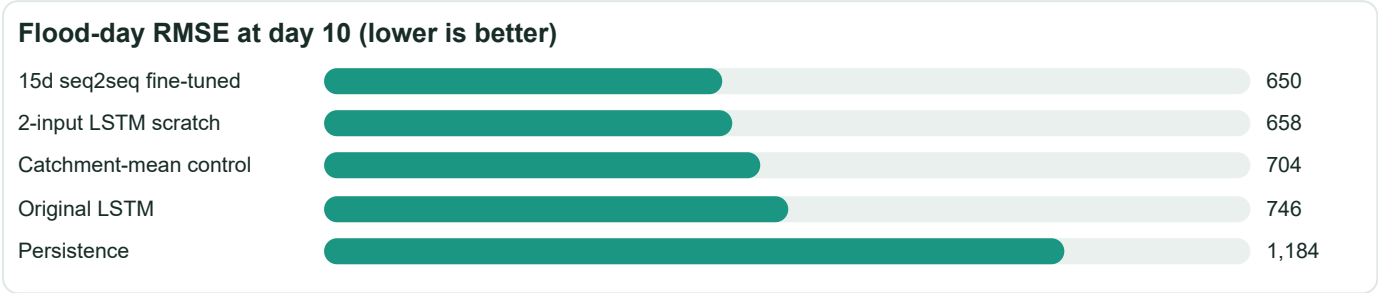
30-day model	Why we tried it	How it makes the forecast	Observed test result at day 30
Weather-only hindcast–forecast LSTM	Baseline neural sequence method without past flow.	Reads 365 days of weather/snow; decoder outputs 30 daily values.	NSE 0.784; flood recall 0.451.
Past-discharge LSTM	Recent river state may add useful memory.	Adds observed flow history to the same weather LSTM.	NSE 0.788; flood recall 0.471; only +0.004 NSE vs weather-only.
Climatology-residual LSTM ensemble	Seasonal average is a tough baseline; model learns deviations around it.	Averages three LSTMs' corrections to day-of-year climatology.	NSE 0.858; best day-30 overall NSE in the listed 30-day table; flood recall 0.333.
15-to-30 fine-tuned encoder–decoder	Tests if a model already trained for 15 leads can extend to 30.	Copies learned 15-day parts, adds 15 leads, fine-tunes three seeds.	NSE 0.849; flood recall 0.412.
delta-HBV	Combines an LSTM with water-balance equations for a snow-fed catchment.	LSTM learns select HBV parameters; physical equations route runoff to flow.	Overall accuracy metrics are in saved comparison; flood-day RMSE is relatively high in the 30-day class report.
Calibrated physical HBV	Provides a physics-based benchmark that can work with limited flow data.	Calibrated snow, soil-moisture, runoff and routing equations.	Day-30 flood recall 0.314; flood RMSE 943 m³/s.
HBV + LSTM correction	Tests whether a neural correction repairs systematic physical-model errors.	LSTM predicts the residual after HBV's flow estimate.	Flood recall 0.745; flood RMSE 559 m³/s. High recall, but also a high false-alarm ratio.
HBV + LightGBM correction	Tree-based correction can learn nonlinear residual patterns with less data.	Lead-specific LightGBM models adjust HBV output.	Flood recall 0.529; flood RMSE 754 m³/s.
Elevation-band LSTM and catchment-mean control	Tests whether spatially distinct high/low basin weather adds useful snow/rain information.	Elevation version learns from band inputs; control learns from catchment-average inputs.	Day-30 NSE: 0.801 vs 0.850. Here the simple mean control did better.
N-HiTS	Tests a multi-scale neural time-series architecture.	Builds forecasts from trend and seasonal blocks at several time scales.	NSE 0.678; flood recall 0.196; below climatology NSE 0.792.
N-BEATSx	Tests a neural basis expansion model that also accepts external weather.	Stacks learned basis blocks to build each horizon's forecast.	NSE 0.564; flood recall 0.451; below climatology.

> A physics model encodes how water can move and be stored. A machine-learning model finds patterns in examples. A hybrid tries to benefit from both. "Why it could work" is a reason to test it; it is not proof that it improved this basin's forecast.

Section 2 · Performance comparison

10-day lead: flood-day error and ordinary-day skill

Common 2022–2024 observed targets. High-flow proxy: discharge > 2,494.7 m³/s (training Q95). There were 51 high-flow targets and 1,024 other targets. Models ranked below by flood RMSE; lower is better. Ordinary-day NSE is shown as a separate check.



Model, lead 10	Flood days	Flood RMSE m³/s	Flood recall	Non-flood NSE
15-day seq2seq fine-tuned	51	649.8	0.804	0.762
15-day seq2seq scratch	51	673.4	0.804	0.765
8-basin LSTM fine-tuned	51	643.0	0.765	0.748
2-input LSTM scratch	51	657.6	0.745	0.745
Global LSTM fine-tuned	51	706.5	0.647	0.861
Snow-bucket + LSTM	51	730.2	0.627	0.767
Original LSTM	51	745.6	0.588	0.837
LSTM + boosting ensemble	51	748.3	0.529	0.852
Matched past-flow/weather LSTM	51	767.8	0.451	0.879
Gradient boosting	51	811.5	0.314	0.840
Elevation-band LSTM	51	896.7	0.196	0.862
Climatology	51	973.6	0.255	0.777
N-HiTS	51	1,091.6	0.471	0.668
N-BEATSx	51	1,229.0	0.275	0.730
Global zero-shot / CAMELS donor-only	51	2,857 / 2,714	0.000 / 0.000	−1.005 / −0.917

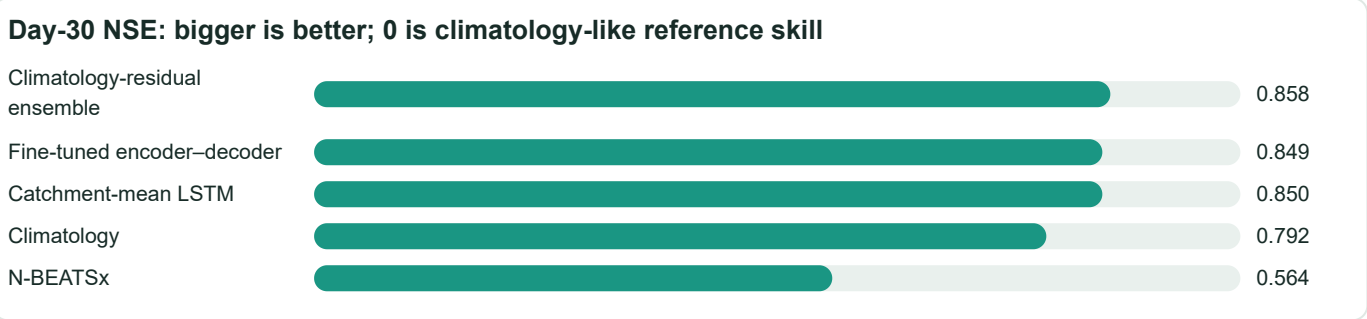
Flood RMSE for all rows is available in the underlying report. Flood recall entries above come from the saved alert-threshold table. Elevation-band and catchment-mean models are shown separately; the latter caught more floods here. Unadapted donor rows are transfer checks only.

> Flood sample is small: 51 high-flow days. Several models catch many floods but issue many false alarms. Neither “best” nor this ranking should be treated as confirmed operational skill.

Section 2 · Longer-lead comparisons

30-day: overall flow score vs high-flow detection

Same observed day-30 test set for the main models: 1,055 targets (51 high-flow and 1,004 non-flood). The table shows that best average-flow score and best flood catch rate are different goals.



30-day model	Day-30 NSE	Flood precision	Flood recall	Flood RMSE m³/s
Climatology-residual LSTM ensemble	0.858	0.548	0.333	826
Catchment-mean LSTM control	0.850	0.491	0.529	728
Fine-tuned encoder–decoder ensemble	0.849	0.382	0.412	802
Calibrated physical HBV	—	0.485	0.314	943
Weather-only LSTM	0.784	0.451	0.451	931
Past-discharge LSTM	0.788	0.490	0.471	904
HBV + LSTM residual	—	0.432	**0.745**	**559**
HBV + LightGBM residual	—	0.435	0.529	754
Spatial elevation-band LSTM	0.801	0.392	0.392	854
N-HiTS	0.678	0.250	0.196	1,293
N-BEATSx	0.564	0.267	0.451	1,164

"—" means the saved comparison table did not report a matching overall NSE for that exact row. Do not combine scores from different cohorts into one ranking.

15-day

The directly comparable fine-tuned vs scratch 15-day encoder–decoder evaluation used the same 2022–2024 benchmark and 51 flood days. At lead 15: fine-tuned flood RMSE 633 m³/s vs scratch 680; flood recall at lead 10 was 0.804 for both. Ordinary-day NSE at lead 10 was 0.762 (fine-tuned) and 0.765 (scratch). Fine-tuning helped the flood error slightly, not every metric.

> **Simple takeaway:** for average flow at 30 days, the residual ensemble scored highest in the saved table. For catching 30-day high flows, HBV + LSTM had the highest recall, but it also had a high false-alarm ratio (about 0.56). For the short 10-day task, the 15-day fine-tuned sequence model and 8-basin fine-tuning show promising flood errors/recall, but these need fresh operational-forecast tests.

Section 3 · Data used for training

Marala target and local inputs

Data	What it measures	Source and role
Daily Marala discharge, q_cumecs	Observed river flow in cubic metres per second; the forecast target.	WAPDA prioritized, with quality-screened FFD bulletin observations supplementing it in the merged daily table. No missing target values were invented for model training; missing targets are masked.
CHIRPS precipitation	Daily rainfall estimate over the catchment.	Primary rainfall input in the merged table; CHIRPS v3. Historical analysis input.
ERA5-Land rainfall	Reanalysis rainfall used in some legacy/model future-weather channels.	Google Earth Engine / ECMWF ERA5-Land. Historical input or proxy for future weather; not an archived forecast.
Temperature	Daily near-surface air temperature.	Google Earth Engine / ECMWF ERA5-Land; supports snowmelt and runoff timing.
Snow water equivalent (SWE)	Water stored in snow.	ERA5-Land-derived daily catchment data.
Snow depth and snowmelt	Snow storage and daily melt.	ERA5-Land-derived daily catchment data.
Dewpoint	Moisture in the air.	ERA5-Land-derived; optional weather input.
Surface solar radiation	Energy from the sun; relates to heating and melt.	ERA5-Land-derived; optional weather input.
Potential evaporation (PET)	Potential water loss from land to air.	ERA5-Land-derived daily estimate; used by hydrology models and some ML inputs.
Elevation-band weather grids	Weather summarized separately by catchment elevation band.	ERA5-Land-derived data used by the spatial LSTM tests.
CAMELS-US donor basins	Observed flow and weather from eight US basins.	Separate multi-basin pretraining data for a transfer-learning experiment; not local Chenab observations.
Nearby gauges / GRRR / GloFAS	Other river discharge series or simulated flow products.	Used in separate reconstruction or diagnostic experiments. These are not treated as WAPDA/FFD observed Marala targets.

Time split and missing data

2016–2020

Training

Fit model weights and training-only input normalization. Five years.

2021

Validation

Choose training stop point and validation adjustments.

2022–2024

Observed test

Score forecasts only when the target discharge was observed.

Many LSTM models used up to 365 previous days and predicted 10, 15, or 30 future days. The newer local neural models mask missing discharge targets rather than filling them. Input gaps may be handled using training-period means and missing-value flags. Weather “forecast” inputs in most experiments were realized reanalysis with added noise because archived 10–30-day weather forecasts were not found.

> **Data limits:** local discharge covers about nine years in the core model table, high-flow test days are scarce, and flow is affected by river regulation. The chosen high-flow threshold is a training-data Q95 proxy. More

> independently held-out years, additional basins, official flood labels, and archived forecast weather would make future evaluations more convincing.

Appendix · Saved horizon inventory and cautions

10-day folder inventory

catchment_mean_lstm · climatology · delta_hbv · ensemble · fine_tuned_global_pretrained_lstm · gradient_boosting · hbv_lightgbm · hbv_lstm · hbv_residual · hybrid_physics_ml · lstm · lstm_past_discharge_weather · lstm_two_input_scratch · N-BEATSx · N-HiTS · persistence · pretrained_multibasin_lstm · pretrained_multibasin_lstm_finetuned · spatially_distributed_model.

15-day folder inventory

discharge_history_lstm · encoder_decoder_lstm · encoder_decoder_lstm_finetuned · grrr_lstm · grrr_lstm_weighted_fills_rejected.

30-day experiments

Weather-only hindcast–forecast LSTM · past-discharge LSTM · climatology-residual LSTM ensemble · fine-tuned 15-to-30 encoder–decoder · delta-HBV · calibrated physical HBV · HBV + LSTM residual · HBV + LightGBM residual · spatial elevation-band LSTM · catchment-mean control · N-HiTS · N-BEATSx.

What not to conclude

- NSE is a skill score, not “percent accurate.”
- Overall accuracy can hide missed floods because ordinary-flow days dominate.
- The high-flow cutoff is a statistical proxy, not a government warning level.
- One small test sample cannot prove that a model will work in future flood seasons.
- Realized reanalysis with artificial noise is not the same as weather information available at forecast time.
- Scores are only directly comparable when dates, targets, lead, threshold and evaluation method match.
- The 30-day fine-tuned encoder result was adjusted using validation information; its interval coverage is not an independent guarantee.

Project records used

Model index: *riverine flood/short range/model/README.md*. Evaluation sources: *evaluation comparison/flood_nonflood_2026-11-02_with_nhits_nbeatsx/report.md*, *evaluation comparison/model_30_days/README.md*, *evaluation comparison/model_30_days/flood_nonflood_classification_metrics.md*, pretrained LSTM, spatial LSTM, encoder–decoder, and 30-day family reports. All results here reflect saved project tables; no models were trained for this document.

> **Practical next step:** keep the observed-only baseline and climatology comparison, then test the leading candidate using archived forecast products that were genuinely available on each historical issue date. Report flood recall, false alarms, peak-flow error, and ordinary-flow scores separately.